

NETWORKING OVERVIEW

FOR CYBERSECURITY BEGINNERS



Understanding how networks work is the foundation of **cybersecurity**.



Learn | Build | Secure



Defend. Detect. Respond. @manishburdak45

1. WHAT IS A NETWORK?



A network is a collection of devices connected together to **communicate** and **share resources**.

EXAMPLES



Laptop accessing the Internet



PC communicating with a Server



Phone connecting to Wi-Fi

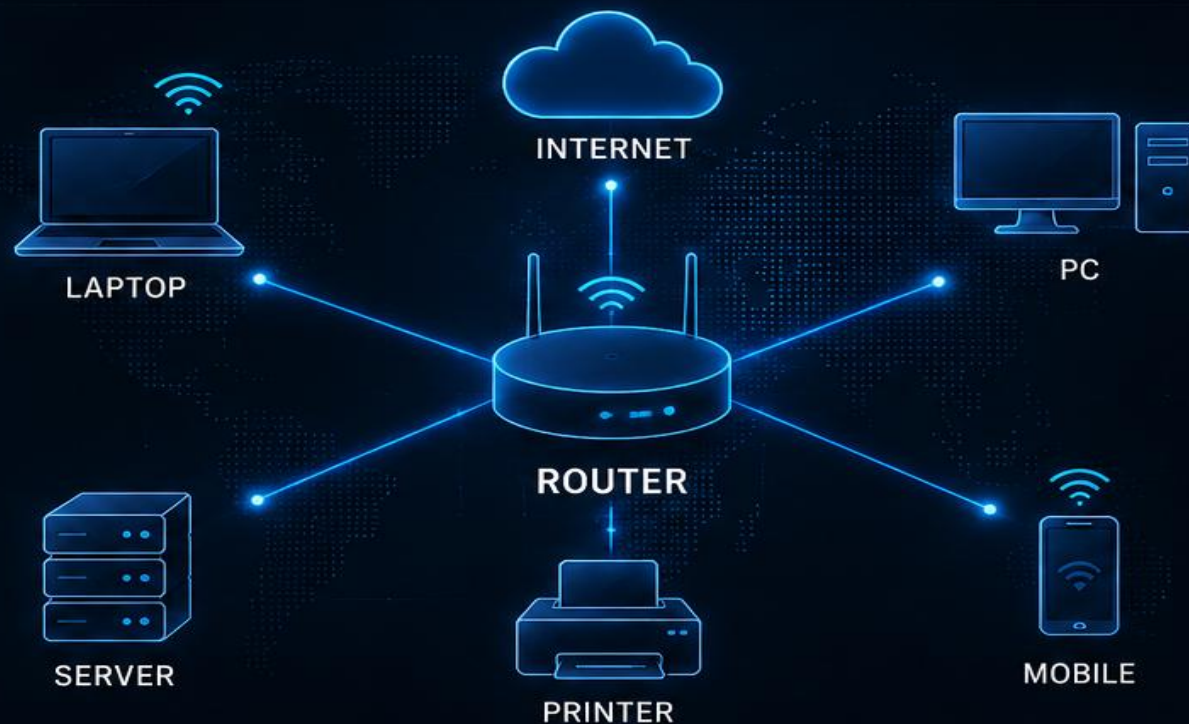
KEY COMPONENTS

- Devices (PCs, Servers, Printers, Phones, etc.)
- Transmission Medium (Ethernet, Fiber, Wireless, etc.)
- Network Devices (Router, Switch, Firewall, etc.)
- Protocols (Rules that define how data is communicated)



For cybersecurity professionals, understanding networks is the **foundation** for everything.

HOW DEVICES COMMUNICATE



DATA FLOW



Data is broken into packets, transmitted through the network, and reassembled at the destination.



KEY TAKEAWAY

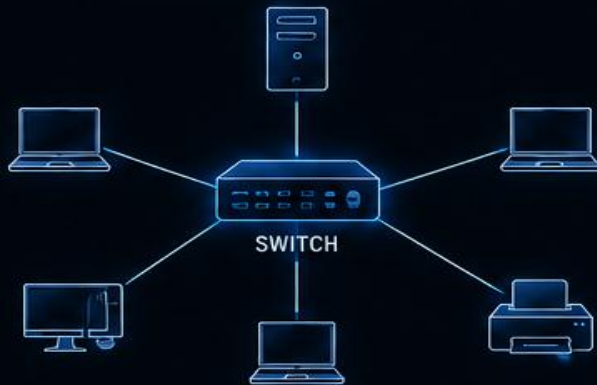
Networks enable **communication**. Secure networks enable **protection**.



2. NETWORK TOPOLOGIES

Topology defines **how devices are connected** in a network.

1. STAR TOPOLOGY



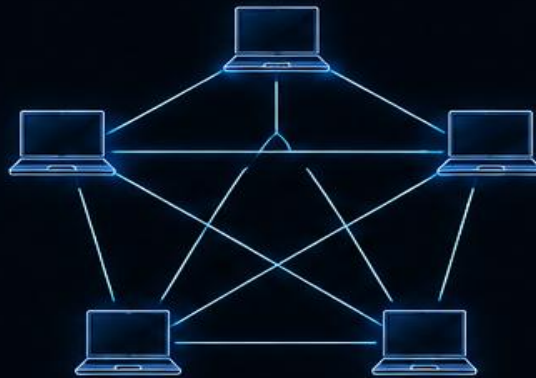
- All devices connect to a central device (switch or hub).
- Easy to manage and troubleshoot.
- Failure of one device doesn't affect others.
- If the central device fails, the entire network goes down.



COMMONLY USED IN:

Home networks, Offices, Most modern LANs

2. MESH TOPOLOGY



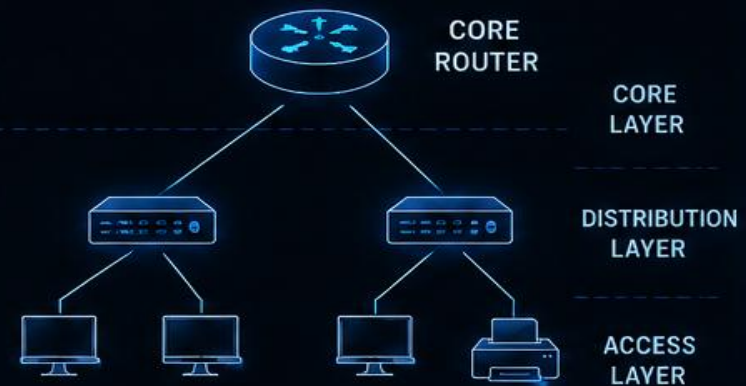
- Every device connects to many or all other devices.
- Very reliable and fault-tolerant.
- Expensive and complex to implement.
- Provides multiple paths for data.



COMMONLY USED IN:

Enterprise backbone, Military, Data centers, Wireless mesh networks

3. TREE TOPOLOGY



- Hierarchical structure (like a tree).
- Scalable and easy to expand.
- Easy management and fault isolation.
- Higher-level device failure can affect lower layers.



COMMONLY USED IN:

Large enterprises, Campus networks, Organizations



KEY TAKEAWAY

Choosing the right topology impacts **performance**, **reliability**, **scalability**, and **security**.



3. KEY NETWORK COMPONENTS

Networks are built using different components that work together to enable communication.

COMMON NETWORK COMPONENTS

END DEVICES



SWITCH



ROUTER



FIREWALL



INTERNET



End Devices

Devices that connect to the network.

- PCs
- Laptops
- Phones
- Servers
- Printers
- IoT Devices



Switch

Connects devices within the same network (LAN).

- Forwards data using MAC addresses
- Reduces network congestion



Router

Connects different networks together and routes data.

- Works with IP addresses
- Determines best path for data



Firewall

Monitors and controls incoming and outgoing traffic based on rules.

- Filters traffic using rules
- Protects network from threats



Internet

A global network of interconnected networks.

- Enables access to external resources
- Uses protocols like TCP/IP

WHAT EACH COMPONENT DOES

COMPONENT		FUNCTION
	End Devices	Generate, send, and receive data.
	Switch	Connects devices in the same network and forwards data efficiently.
	Router	Routes data between different networks.
	Firewall	Allows or blocks traffic based on security rules.
	Internet	Provides connectivity to external networks and resources.

EXAMPLE



Data flows from your device to the Internet through multiple components.

CYBERSECURITY RELEVANCE



Attackers target components to gain unauthorized access. Securing each component is essential for a strong defense-in-depth strategy.

KEY TAKEAWAY



Understanding these components helps you design, secure, and troubleshoot networks effectively.

4. FLAT NETWORK vs SEGMENTED NETWORK

Network segmentation divides a network into smaller parts to improve security, performance and control traffic flow.

FLAT NETWORK (NO SEGMENTATION)

All devices are in the same network.
Everyone can communicate with each other.

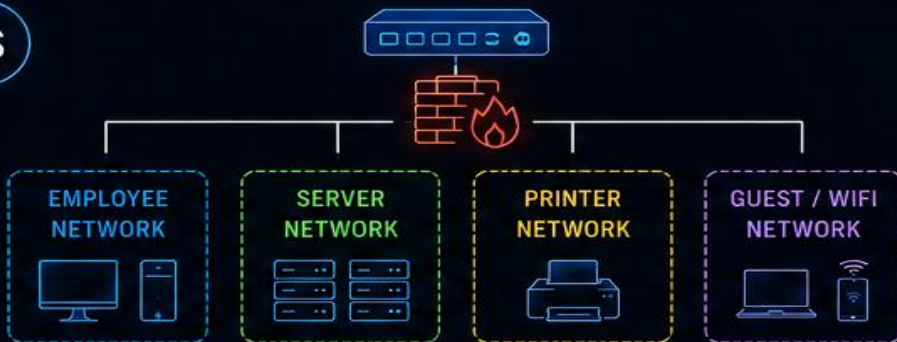


If attacker compromises one device, they can easily access everything.

VS

SEGMENTED NETWORK

Network is divided into smaller parts (segments).
Communication is controlled.



If attacker compromises one device, their access is limited to that segment only.

BENEFITS OF SEGMENTATION



IMPROVED SECURITY

Limits attacker movement and lateral attacks.



BETTER PERFORMANCE

Reduces broadcast traffic and congestion.



ACCESS CONTROL

Apply rules (ACLs) between segments.



EASY MONITORING

Identify issues and threats quickly per segment.



COMPLIANCE

Helps meet security policies and industry standards.

COMPARISON TABLE

FEATURE	FLAT NETWORK	SEGMENTED NETWORK
Structure	Single large network	Multiple smaller networks
Security	Low - Easy for attackers to move	High - Limits attacker movement
Traffic Control	Difficult to control	Easy to control with rules (ACLs)
Performance	More congestion	Optimized traffic flow
Management	Hard to manage	Easier to manage
Failure Impact	One issue affects all	Limited impact to specific segment

EXAMPLE - ENTERPRISE NETWORK



Each department is isolated. Access allowed only as needed.



PRACTICAL TIP (KALI / LINUX)

You can check your IP, route and network info using these commands:

```
ip a           # Show IP addresses
ip route       # Show routing table
ping <target_ip> # Test connectivity
nmap <target_ip> # Scan open ports
ss -tln       # Show listening ports
```



KEY TAKEAWAY:

Flat networks are easy but risky. Segmentation is secure, controlled and enterprise-ready.

5. IP ADDRESSING & SUBNETTING

IP addressing allows devices to **identify** and **communicate** with each other on a network.

IP ADDRESS STRUCTURE (IPv4)

An IPv4 address is 32 bits (4 bytes) written in decimal.

192 . 168 . 1 . 10

1 Byte (8 bits) 1 Byte (8 bits) 1 Byte (8 bits) 1 Byte (8 bits)

Network Part

Host Part

Example:

IP Address : 192.168.1.10
Subnet Mask : 255.255.255.0 (/24)
Network : 192.168.1.0
Broadcast : 192.168.1.255
Usable Hosts : 192.168.1.1 – 192.168.1.254

Binary (32 bits)

11000000 . 10101000 . 00000001 . 00001010

Network

Host

SUBNETTING EXAMPLES

CIDR	Subnet Mask	No. of Networks	No. of Hosts per Network
/24	255.255.255.0	1	254
/25	255.255.255.128	2	126
/26	255.255.255.192	4	62
/27	255.255.255.224	8	30
/28	255.255.255.240	16	14
/29	255.255.255.248	32	6
/30	255.255.255.252	64	2



Why Subnet?

- Divide a large network into smaller, manageable parts.
- Improve security, performance, and reduce broadcast traffic.
- Enable efficient IP address utilization.

SUBNETTING VISUAL (192.168.1.0/24)

Original Network: 192.168.1.0/24
(254 Hosts)



192.168.1.0/24

192.168.1.0/25
(126 Hosts)

192.168.1.128/25
(126 Hosts)



Smaller subnets = Better control & security

PRIVATE IP ADDRESS RANGES

These IP ranges are used in private networks and are **not** routable on the Internet.



Class A	10.0.0.0	– 10.255.255.255	(/8)
Class B	172.16.0.0	– 172.31.255.255	(/16)
Class C	192.168.0.0	– 192.168.255.255	(/24)



Used in homes, offices, organizations.
Requires NAT to access the Internet.



COMMON COMMANDS (KALI / LINUX)

```
ip a           # Show IP addresses
ip route       # Show routing table
ifconfig       # Show network interfaces (legacy)
ping <ip>      # Test connectivity
netstat -tuln  # Show listening ports
nmap <ip>      # Scan open ports
traceroute <ip> # Trace route to destination
arp -a         # Show ARP table (local network)
```



CYBERSECURITY RELEVANCE



IP addressing helps in reconnaissance and enumeration.
(e.g., **subnet discovery**, **host discovery**)



Subnetting helps in **network segmentation**, **access control**, and limiting attacker movement.



Understanding IP and routing is critical for **penetration testing**, **incident response**, and **threat detection**.



KEY TAKEAWAY

IP addressing and subnetting are the **foundation** of all network communication. Mastering these concepts is essential for both **networking** and **cybersecurity**.

6. PROTOCOLS – THE LANGUAGE OF NETWORKS

Protocols are rules that define how data is formatted, transmitted, received, and understood between devices.

COMMON NETWORK PROTOCOLS



TCP
Transmission Control Protocol

- Connection-oriented
- Reliable & error-checked
- Data delivered in order
- Uses 3-way handshake

Used For:

 Web (HTTPS)  SSH  Email (SMTP)  File Transfer (FTP)



UDP
User Datagram Protocol

- Connectionless
- Faster but not reliable
- No guarantee of delivery
- No handshake

Used For:

 Video Streaming  Online Gaming  VoIP / Calls  DNS Queries



IPX
Internetwork Packet Exchange

- Older protocol (NetWare)
- Now mostly obsolete
- Used for internal network communication

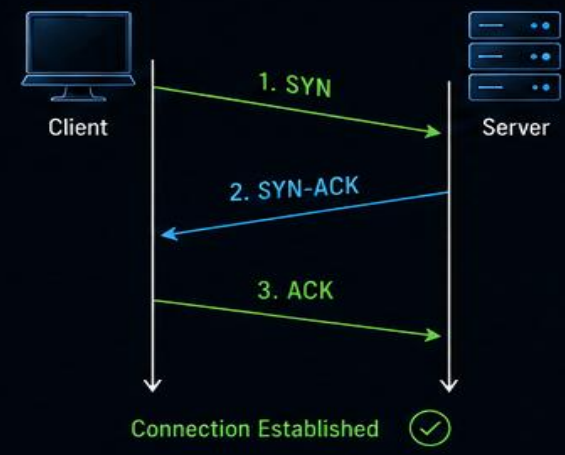
Used For:

 Legacy systems and older Novell NetWare environments

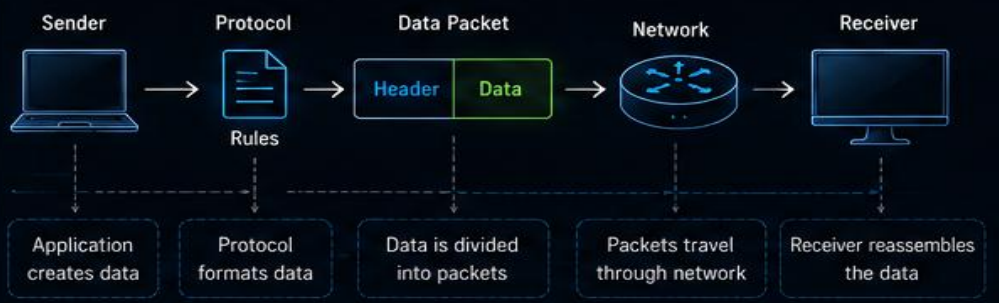
TCP vs UDP COMPARISON

FEATURE	TCP	UDP
Connection	Connection-oriented	Connectionless
Reliability	Reliable	Not Reliable
Speed	Slower	Faster
Error Handling	Yes	No
Handshake	Yes (3-way)	No
Use Case	Critical Data	Real-time Data

TCP 3-WAY HANDSHAKE



HOW DATA IS COMMUNICATED



EXAMPLE – WEB BROWSING (HTTPS)



CYBERSECURITY RELEVANCE

- Attackers abuse protocols (e.g., TCP SYN Flood DDoS, DNS Amplification using UDP).
- Understanding protocols helps in traffic analysis, intrusion detection and malware analysis.
- Wrong protocol configurations can lead to data leakage or exploitation.
- Monitoring unusual protocol behavior is key to detecting attacks early.

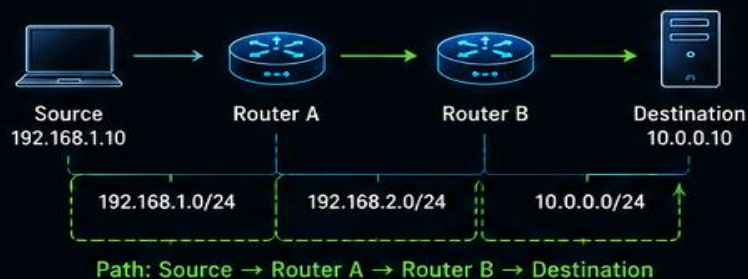
7. ROUTING & GATEWAYS – PATHS TO DESTINATION

Routing allows data to travel from the **source** to the **destination** across multiple networks.

Gateways act as the **exit** or **entry point** between different networks.

WHAT IS ROUTING?

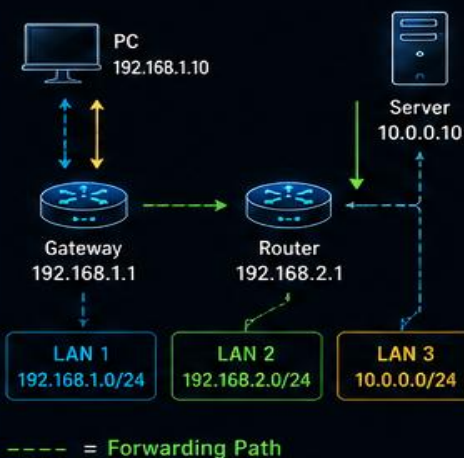
Routing is the process of selecting the best path for data to travel from source to destination across multiple networks.



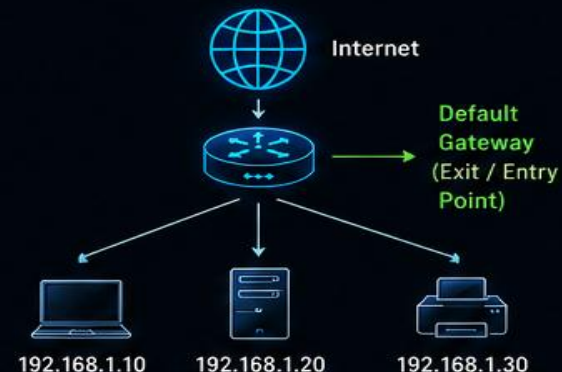
Routers use routing tables to decide the next hop towards the destination.

HOW ROUTING WORKS

- 1 Source sends data to its default gateway.
- 2 Gateway checks routing table to find the best path.
- 3 Packet is forwarded to the next hop router.
- 4 This continues until the destination network is reached.
- 5 Packet is delivered to the destination.



GATEWAYS EXPLAINED



Without a gateway, devices in a local network cannot communicate with external networks.

ROUTING TABLE EXAMPLE (Router A)

Destination Network	Subnet Mask	Next Hop	Interface	Metric
192.168.1.0	255.255.255.0	Directly Connected	192.168.1.1	0
192.168.2.0	255.255.255.0	Directly Connected	192.168.2.1	0
10.0.0.0	255.255.255.0	192.168.2.2	192.168.2.1	1
0.0.0.0	0.0.0.0	192.168.1.1	192.168.1.1	1



0.0.0.0/0 is the default route (gateway of last resort).

PRACTICAL LAB (KALI / LINUX)

```
> # Show IP addresses
ip a
# Show routing table
ip route
# Add a static route (example)
sudo ip route add 10.0.0.0/24 via 192.168.2.1
# Delete a static route
sudo ip route del 10.0.0.0/24 via 192.168.2.1
# Test connectivity
ping 10.0.0.10
```



CYBERSECURITY RELEVANCE



Attackers modify routes (e.g., rogue router, route hijacking) to intercept traffic.



Improper routing can expose internal networks to the Internet.



Monitoring routing tables helps detect suspicious changes.



Understanding gateways is critical for pivoting and lateral movement during pentesting.



KEY TAKEAWAY

Routing determines the **path**, and **gateways** connect networks.

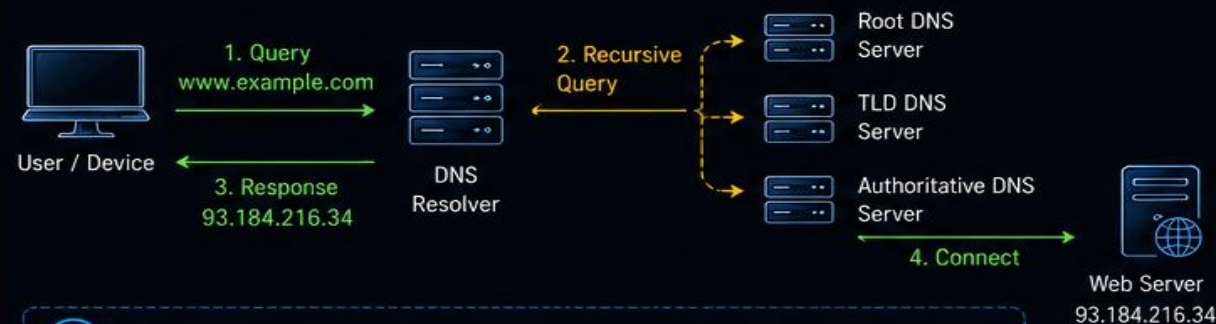
A solid understanding of routing is essential for both network **troubleshooting** and **cybersecurity**.

8. DNS, URL vs FQDN – NAMING & RESOLUTION

DNS translates human-friendly names into IP addresses. URLs are complete web addresses, while FQDNs identify hosts in a network.

DNS – THE INTERNET PHONEBOOK

DNS (Domain Name System) converts domain names into IP addresses so that devices can find and communicate with each other.



i DNS happens in the background every time you open a website or use an online service.

COMMON DNS RECORD TYPES

RECORD TYPE	PURPOSE	EXAMPLE
A	Maps domain to IPv4 address	example.com → 93.184.216.34
AAAA	Maps domain to IPv6 address	example.com → 2606:2800:220:1:248:1893:25c8:1946
CNAME	Alias of another domain	www.example.com → example.com
MX	Mail exchange server	example.com → mail.example.com
NS	Authoritative name server	example.com → ns1.example.com
TXT	Text information (SPF, verification, etc.)	example.com → "v=spf1 include:_spf.example.com ~all"

URL vs FQDN

FQDN (FULLY QUALIFIED DOMAIN NAME)

Specifies the exact domain name of a host.

www.example.com

Components:

- Hostname: www
- Domain: example
- Top-Level Domain: .com

No protocol, path or parameters.

URL (UNIFORM RESOURCE LOCATOR)

Complete address to access a resource on the web.

https://www.example.com/login?user=1

Components:

- Protocol: https://
- Domain (FQDN): www.example.com
- Path: /login
- Query/Parameters: ?user=1

Includes protocol, path and parameters.

EXAMPLE WORKFLOW

- User types: `https://www.example.com/login`
- DNS resolves `www.example.com` to IP: `93.184.216.34`
- Browser connects to `93.184.216.34` using HTTPS (port 443)
- Web server returns the requested page (`/login`) to the browser

CYBERSECURITY RELEVANCE

- DNS is a common target for attacks like cache poisoning, tunneling, and DDoS.
- Monitoring DNS traffic helps detect malicious domains and data exfiltration.
- Understanding URLs helps in identifying phishing attempts and malicious links.
- FQDNs are used in logs, certificates, firewall rules, and threat intelligence.

KEY TAKEAWAY

DNS helps us find the right **destination**, FQDNs **uniquely identify** hosts, and URLs guide us to the **exact resource**. Understanding these is essential for **networking** and **cybersecurity**.

9. PORTS & SERVICES – THE DOORS OF COMMUNICATION

Ports identify specific processes or services on a device. They help data reach the right application.

WHAT IS A PORT?

A port is a logical endpoint where network connections start or end. Each port is represented by a number from 0 to 65535.



i IP address identifies the device, port identifies the application/service.

PORT NUMBER RANGES

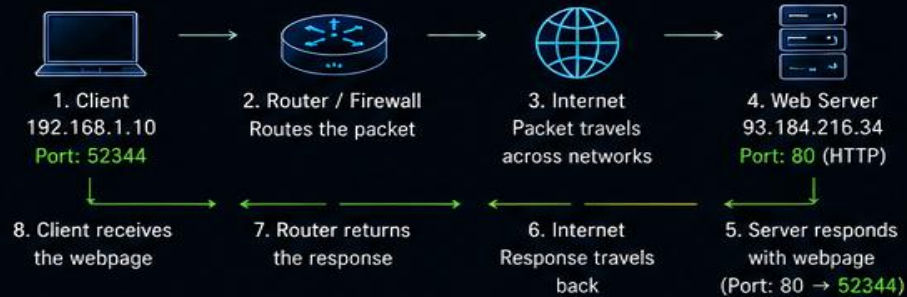
0 – 1023	Well-Known Ports	Assigned to common services and protocols.
1024 – 49151	Registered Ports	Assigned to applications by organizations.
49152 – 65535	Dynamic / Ephemeral Ports	Used for temporary connections (client side).

i Well-known ports are the most important for networking and troubleshooting.

COMMON WELL-KNOWN PORTS

PORT	PROTOCOL	SERVICE	DESCRIPTION
20 / 21	TCP	FTP	File Transfer Protocol
22	TCP	SSH	Secure Shell (Remote Login)
23	TCP	Telnet	Remote Terminal Access
25	TCP	SMTP	Simple Mail Transfer Protocol
53	UDP/TCP	DNS	Domain Name System
67 / 68	UDP	DHCP	Dynamic Host Configuration
80	TCP	HTTP	HyperText Transfer Protocol
110	TCP	POP3	Post Office Protocol v3
143	TCP	IMAP	Internet Message Access Protocol
443	TCP	HTTPS	HTTP Secure (SSL/TLS)
3389	TCP	RDP	Remote Desktop Protocol

HOW PORTS WORK (EXAMPLE – WEB BROWSING)



i The client uses a temporary port (ephemeral) to communicate with a well-known port on the server.

COMMON SERVICES & THEIR PORTS

Web (HTTP) Port 80 (TCP)	SSH Port 22 (TCP)
Secure Web (HTTPS) Port 443 (TCP)	FTP Port 20/21 (TCP)
Email (SMTP) Port 25 (TCP)	Remote Desktop (RDP) Port 3389 (TCP)
DNS Port 53 (UDP/TCP)	DHCP Port 67/68 (UDP)

💡 Memorizing common ports helps in troubleshooting and securing your network.

CYBERSECURITY RELEVANCE

- Unnecessary open ports can be exploited by attackers.
- Use firewalls to allow only required ports and block the rest.
- Port scanning helps attackers find open ports and potential entry points.
- Services running on default ports are more predictable and risky.
- Changing non-standard ports can add an extra layer of security (security by obscurity).



PRACTICAL LAB (KALI / LINUX)

```
# Check open ports on your system
ss -tln
# Scan open ports on a target
nmap 192.168.1.10
# Scan top 1000 ports
nmap -sS -p --top-ports 1000 192.168.1.10
```



Useful tool

nmap – Network scanning tool
ss / netstat – View open ports
telnet <ip> <port> – Test connection
nc <ip> <port> – Netcat connection test

QUICK REFERENCE – PORT ACCESS EXAMPLE

SCENARIO	COMMAND	EXAMPLE
Check if HTTP (80) is open	nc -zv 93.184.216.34 80	nc -zv example.com 80
Check if SSH (22) is open	nc -zv 192.168.1.10 22	nc -zv 192.168.1.10 22
Port scan (top 100)	nmap --top-ports 100 192.168.1.10	nmap --top-ports 100 192.168.1.10



KEY TAKEAWAY:

Ports and services ensure data reaches the right application. Knowing important **ports**, their **functions**, and **securing** them properly is essential for building strong networks and **defending** against attacks.

10. TCP CONNECTION – THE RELIABLE WORKHORSE

TCP (Transmission Control Protocol) provides reliable, ordered, and error-checked delivery of data between applications.

TCP CONNECTION ESTABLISHMENT (3-WAY HANDSHAKE)

Before data can be exchanged, TCP establishes a connection.



- SYN: Client wants to connect.
- SYN-ACK: Server acknowledges and sends its own sequence.
- ACK: Client acknowledges. Connection established!

TCP CONNECTION TERMINATION (4-WAY HANDSHAKE)

Either side can close the connection.



- Each side confirms the other's close request. Ensures all data is delivered before the connection ends.

TCP HEADER FORMAT



IMPORTANT FIELDS

- **Source/Destination Port:** Identify sending and receiving applications.
- **Sequence/Acknowledgment:** Ensure ordered and reliable delivery.
- **Flags:** Control the connection (SYN, ACK, FIN, RST, PSH, etc.).
- **Window Size:** Flow control – how much data the receiver can accept.

TCP RELIABILITY FEATURES



Reliable Delivery

Uses acknowledgments (ACKs) and retransmissions.



Ordered Delivery

Sequence numbers ensure data is reassembled in order.



Error Detection

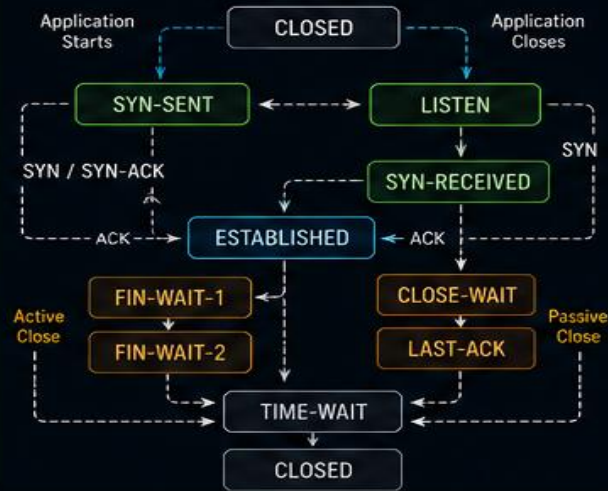
Checksum detects corrupted segments.



Flow Control

Uses window size to prevent receiver overload.

TCP STATES (SIMPLIFIED)



COMMON TCP FLAGS

FLAG	NAME	PURPOSE
SYN	Synchronize	Initiate connection
ACK	Acknowledgment	Acknowledge received data
FIN	Finish	Close connection
RST	Reset	Abort connection immediately
PSH	Push	Push data to the application
URG	Urgent	Urgent pointer is valid

CYBERSECURITY RELEVANCE



SYN Flood Attacks

Attackers send many SYN packets to exhaust server resources.



Session Hijacking

Stealing TCP session data allows unauthorized access.



TCP Reset Attacks

RST packets can be abused to drop legitimate connections.



Monitoring TCP Behavior

Analyzing flags, states, and patterns helps detect anomalies.



KEY TAKEAWAY

TCP ensures **reliable**, **ordered**, and **secure** communication through connection management, flow control, and error handling. Understanding TCP is essential for **networking**, **troubleshooting**, and **defending** against attacks.

11. UDP – FAST BUT CONNECTIONLESS

UDP (User Datagram Protocol) is a connectionless, lightweight transport protocol that prioritizes speed over reliability. It is ideal for real-time applications where occasional data loss is acceptable.

WHAT IS UDP?

- Connectionless – No handshake
- No reliability – No ACKs, retransmissions, or guaranteed delivery
- Message-oriented – Preserves message boundaries
- Low overhead – Smaller header, faster processing
- Best-effort delivery – Packets may be lost, duplicated, or arrive out of order

UDP vs TCP (Quick Comparison)

Feature	TCP	UDP
Connection	Connection-oriented	Connectionless
Reliability	Reliable (ACKs, retransmit)	Not Reliable
Order	In-order delivery	No order guarantee
Speed	Slower	Faster
Overhead	Higher	Lower
Use Case	Web, Email, File Transfer	Streaming, Gaming, VoIP, DNS

UDP HEADER FORMAT

0	15	16	31
Source Port (16 bits)		Destination Port (16 bits)	
Length (16 bits)			
Checksum (16 bits)			
Data (Variable Length)			

Source Port: Identifies the sending application.
Destination Port: Identifies the receiving application.
Length: Total length of UDP header and data.
Checksum: Error-checking of header and data (optional in IPv4).

UDP COMMUNICATION EXAMPLE

1. Send Request (No Handshake)
2. Send Response (No ACK Required)

Client: 192.168.1.10
Server: 192.168.1.20

UDP Packet

Info: No connection setup or teardown. Data is sent immediately. No ACKs, no retransmissions.

HOW UDP WORKS

1. Application creates data
2. UDP adds header
3. Datagram is sent over IP
4. UDP removes header and delivers data to application

Info: Each UDP datagram is independent. Packets may take different paths, be delayed, duplicated, or lost.

COMMON USES OF UDP

Video Streaming

Online Gaming

VoIP / Video Calls

DNS Queries

Live Broadcasts

IoT / Sensor Data

Network Monitoring

Online Speed Tests

Info: These applications value speed and low latency over perfect delivery.

ADVANTAGES

- ✓ Very fast and efficient
- ✓ Low latency
- ✓ Low overhead (8-byte header)
- ✓ Ideal for real-time communication
- ✓ Supports broadcast and multicast

DISADVANTAGES

- ✗ No guarantee of delivery
- ✗ No order guarantee
- ✗ No flow control
- ✗ No error recovery
- ✗ Not suitable for important data transfers

UDP vs TCP HEADER SIZE

TCP Header

20–60 bytes

UDP Header

8 bytes

Info: Smaller header = less overhead = faster transmissions.

WHEN TO USE UDP?

- ✓ Real-time applications where small delays matter
- ✓ Streaming audio/video where some loss is tolerable
- ✓ Games where responsiveness is critical
- ✓ DNS queries and lookups

- ✗ When every byte must arrive (use TCP)
- ✗ File transfers
- ✗ Email
- ✗ Web browsing (HTTP/HTTPS)

CYBERSECURITY RELEVANCE

- ⚠ UDP is used in attacks like UDP Flood, DNS Amplification, NTP Amplification, and DDoS.
- 🔍 Harder to mitigate because UDP is connectionless and stateless.
- 🛡 Monitoring unusual UDP traffic and open UDP ports is important for security.
- 👁 Filtering and rate limiting can help defend against malicious UDP traffic.

KEY TAKEAWAY

UDP is the “speedy” protocol of the Internet. It is perfect for real-time, low-latency applications but doesn’t guarantee delivery. Use UDP when **speed** matters more than **reliability**.



13. IP ADDRESSING – THE LANGUAGE OF NETWORKS

IP addresses uniquely identify devices on a network and enable communication across networks.

WHAT IS AN IP ADDRESS?

An IP address is a logical address assigned to a device for identification and communication.

IPv4 (32-bit)

Dotted Decimal Notation

192.168.1.10

11000000.10101000.00000001.00001010

IPv6 (128-bit)

Colon Hexadecimal Notation

2001:0db8:85a3:0000:0000:8a2e:0370:7334

IPv4 uses 32 bits (4.3 billion addresses).
IPv6 uses 128 bits (340 undecillion addresses).

IP ADDRESS STRUCTURE (IPv4 EXAMPLE)

Example IP: 192.168.1.10/24

192

168

1

10

Network Portion Host Portion

CIDR	Subnet Mask	Network Address	Usable Host Range	Broadcast Address
/24	255.255.255.0	192.168.1.0	192.168.1.1 – 192.168.1.254	192.168.1.255

CIDR (Classless Inter-Domain Routing) uses /prefix length.
/24 means the first 24 bits are the network portion.

COMMON PRIVATE IPv4 RANGES

CLASS	RFC 1918 RANGE	SUBNET MASK	USABLE HOSTS
Class A	10.0.0.0 – 10.255.255.255	255.0.0.0	16,777,214
Class B	172.16.0.0 – 172.31.255.255	255.255.0.0	1,048,574
Class C	192.168.0.0 – 192.168.255.255	255.255.255.0	65,534

SPECIAL IPv4 ADDRESSES

0.0.0.0	Unspecified address (used during boot)
127.0.0.1	Loopback address (localhost)
169.254.0.0/16	Link-local address (APIPA)
255.255.255.255	Limited broadcast
Network Address	First address in a subnet (e.g., 192.168.1.0)
Broadcast Address	Last address in a subnet (e.g., 192.168.1.255)

SUBNETTING BASICS

Example: 192.168.1.0/24
Need 4 subnets → Borrow 2 bits → /26

Network (24 bits)

Host (8 bits)

11000000101010000000000000000000

/26

Subnet	Network Address	Usable Host Range	Broadcast Address
1	192.168.1.0/26	192.168.1.1 – 192.168.1.62	192.168.1.63
2	192.168.1.64/26	192.168.1.65 – 192.168.1.126	192.168.1.127
3	192.168.1.128/26	192.168.1.129 – 192.168.1.190	192.168.1.191
4	192.168.1.192/26	192.168.1.193 – 192.168.1.254	192.168.1.255

Subnetting divides a network into smaller networks by borrowing bits from the host portion.

IP ADDRESS ASSIGNMENT

Static IP



- Manually configured
- Does not change
- Used for servers, printers, network devices

Dynamic IP (via DHCP)



DHCP

- DHCP Discover
- DHCP Offer
- DHCP Request
- DHCP Acknowledge

- Automatically assigned
- Can change over time
- Used for clients, laptops, phones

DHCP simplifies IP management by automatically assigning and renewing IP addresses.

IPv6 ADDRESS EXAMPLE

Example: 2001:db8:85a3::8a2e:370:7334/64

2001

db8

85a3

0000

0000

8a2e

0370

7334

Network Prefix (/64) Interface ID

IPv6 uses shorthand (::) to compress consecutive zeros.
/64 is the common prefix length for networks.

TIPS & BEST PRACTICES

- Plan your IP scheme before deployment.
- Use private IPs internally, public IPs for Internet.
- Avoid overlapping subnets.
- Reserve IPs for critical devices.
- Document your IP address plan.

QUICK CALCULATION (IPv4)

Usable Hosts = 2^(Host Bits) – 2

Examples:
/24 → 2^(8) – 2 = 254 hosts
/26 → 2^(6) – 2 = 62 hosts

CYBERSECURITY RELEVANCE

- IP spoofing can be used to hide attacker identity.
- Proper IP planning helps with segmentation and reduces attack surface.
- Monitor unusual IP activity and unauthorized devices.
- Use ACLs and firewalls to control IP-based access.

14. MAC ADDRESSING – THE DATA LINK IDENTIFIER

MAC (Media Access Control) addresses are unique hardware identifiers used at the Data Link layer (Layer 2) to deliver frames between devices on the same network.

WHAT IS A MAC ADDRESS?

- Unique identifier assigned to a network interface (NIC).
- Used for local (LAN) communication.
- Operates at OSI Layer 2 (Data Link).
- Hardcoded (burned-in) or locally administered.

Example:
3C:52:82:1A:2B:3C
OUI (24 bits) identifies the vendor
NIC Specific (24 bits) Assigned by vendor



MAC addresses are written as 6 groups of 2 hexadecimal digits separated by colons (:) or hyphens (-).

MAC ADDRESS TYPES

Type	Bit (U/L)	Bit (I/G)	Description	Example
Unicast	0	0	Addressed to a single interface	3C:52:82:1A:2B:3C
Multicast	0	1	Addressed to a group of interfaces	01:00:5E:00:00:FB
Broadcast	1	1	Addressed to all interfaces on the local network	FF:FF:FF:FF:FF:FF
Locally Administered (Unicast)	1	0	Manually assigned (unicast) by user/admin	02:1A:2B:3C:4D:5E



U/L bit (bit 1 of first octet): 0 = Universal (globally unique), 1 = Local (admin assigned)
I/G bit (bit 0 of first octet): 0 = Unicast, 1 = Multicast/Broadcast

MAC ADDRESS STRUCTURE



Examples of OUI (Vendors)

OUI (Hex)	Vendor	OUI (Hex)	Vendor
00:1A:2B	Cisco Systems	08:00:27	VirtualBox
00:1B:63	Apple, Inc.	00:50:56	VMware (old)
00:0C:29	VMware, Inc.	F0:76:1C	Samsung Electronics
3C:52:82	Dell Inc.	7C:DD:90	Intel Corporate



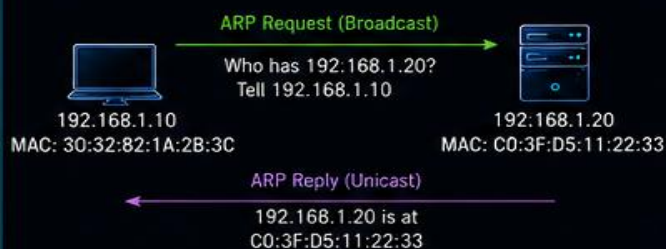
Look up OUIs at: <https://standards-oui.ieee.org/oui>

HOW MAC COMMUNICATION WORKS (SAME LAN)



MAC addresses are used within the local network. Routers are used to reach other networks.

ARP – RESOLVING IP TO MAC



ARP maps IP addresses to MAC addresses and is essential for local network communication.

COMMON MAC ADDRESSES

Address	Purpose
FF:FF:FF:FF:FF:FF	Broadcast (all devices on the LAN)
01:00:5E:xx:xx:xx	IPv4 Multicast (mapped range)
33:33:xx:xx:xx:xx	IPv6 Multicast
00:00:00:00:00:00	Unspecified (invalid source)
00:1A:2B:00:00:00	Example of vendor OUI



Some MACs have special purposes and should be recognized in network troubleshooting.

MAC ADDRESS TABLE (SWITCH CAM TABLE)

MAC Address	Port	VLAN	Type	Aging Time
3C:52:82:1A:2B:3C	Gi0/1	10	Dynamic	200
C0:3F:D5:11:22:33	Gi0/2	10	Dynamic	195
02:1A:2B:3C:4D:5E	Gi0/3	20	Dynamic	180
00:1A:2B:AA:BB:CC	Gi0/1	10	Static	-



Switches learn MAC addresses and build a table to forward frames only to the correct port.

BEST PRACTICES

- Secure ports (Port Security) to limit allowed MAC addresses.
- Monitor for MAC address flapping (a sign of network issues or attacks).
- Use static MAC entries for critical devices.
- Keep firmware updated to prevent MAC table overflow attacks.
- Segment networks properly to reduce unnecessary broadcasts.

CYBERSECURITY RELEVANCE



MAC Spoofing: Attackers can spoof MAC addresses to bypass access controls.



MAC Flooding: Overloads the switch MAC table, causing performance issues.



Port Security and 802.1X: Help prevent unauthorized devices from connecting.



Monitoring MAC activity helps detect abnormal behavior and potential threats.



KEY TAKEAWAY:

MAC addressing is the foundation of **local network** communication. Understanding its structure, types, and usage is crucial for **networking** efficiency, **troubleshooting**, and **network security**.

15. SUBNETTING – SCALING NETWORKS EFFICIENTLY

15

Subnetting divides a larger network into smaller, manageable sub-networks to improve performance, security, and IP address utilization.

WHY SUBNET?

- ✓ Efficient use of IP addresses
- ✓ Improved performance (smaller broadcast domains)
- ✓ Enhanced security (isolation between departments/VLANs)
- ✓ Simpler network management
- ✓ Supports route summarization

i Subnetting borrows bits from the host portion to create subnets.

IP ADDRESS & SUBNET MASK BASICS

Example IP: 192.168.1.10 / 24



/24 (255.255.255.0)

Subnet mask defines how many bits belong to the network portion.
/24 mask = 24 network bits, 8 host bits.

i More network bits = more subnets, fewer hosts per subnet.
Fewer network bits = fewer subnets, more hosts per subnet.

SUBNETTING PROCESS (4 STEPS)

- Determine Requirements**
How many subnets and hosts per subnet are needed?
- Borrow Bits**
Borrow host bits to create enough subnets.
- Calculate New Subnet Mask**
Add borrowed bits to the default mask.
- Determine Subnet Details**
Find network address, broadcast address, and usable host range for each subnet.

SUBNETTING EXAMPLE

Given: 192.168.1.0 / 24

Need	Borrow Bits	New Prefix	Subnet Mask	# of Subnets	# of Hosts per Subnet
2 subnets	1	/25	255.255.255.128	2	126
4 subnets	2	/26	255.255.255.192	4	62
8 subnets	3	/27	255.255.255.224	8	30
16 subnets	4	/28	255.255.255.240	16	14
32 subnets	5	/29	255.255.255.248	32	6
64 subnets	6	/30	255.255.255.252	64	2
128 subnets	7	/31	255.255.255.254	128	2 (PtP)

i Use the smallest subnet that meets your requirements to conserve IPs.

SUBNETTING EXAMPLE (DETAILED)

Given: 192.168.1.0 / 24
Need: 4 subnets

Step 1 & 2: Borrow 2 bits
/24 → /26 (255.255.255.192)

Step 3: New Prefix
/26

Subnet	Subnet ID (Binary)	Subnet ID (Decimal)	Network Address	Usable Host Range	Broadcast Address
1	00	0	192.168.1.0	192.168.1.1 – 192.168.1.62	192.168.1.63
2	01	64	192.168.1.64	192.168.1.65 – 192.168.1.126	192.168.1.127
3	10	128	192.168.1.128	192.168.1.129 – 192.168.1.190	192.168.1.191
4	11	192	192.168.1.192	192.168.1.193 – 192.168.1.254	192.168.1.255

i First address = network, last address = broadcast, others = usable hosts.

VLSM – VARIABLE LENGTH SUBNET MASKING

VLSM allows using different subnet sizes within the same network to match different host requirements.

Example: 192.168.1.0 / 24

Department	Hosts Needed	Subnet	Usable Host Range	Broadcast
Engineering	120	/25	192.168.1.1 – 192.168.1.126	192.168.1.127
Sales	60	/26	192.168.1.129 – 192.168.1.190	192.168.1.191
IT	28	/27	192.168.1.193 – 192.168.1.222	192.168.1.223
Guest Wi-Fi	12	/28	192.168.1.225 – 192.168.1.238	192.168.1.239
Management	2	/30	192.168.1.241 – 192.168.1.242	192.168.1.243

i Allocate largest subnets first, then smaller ones.

QUICK CALCULATION TIPS



Total Subnets = $2^{\text{borrowed bits}}$

Usable Hosts per Subnet = $2^{\text{host bits}} - 2$
(subtract network & broadcast)

Host Bits = 32 – Prefix Length

Wildcard Mask = 255.255.255.255 – Subnet Mask

Common Host Counts

i /26 = 62 hosts /27 = 30 hosts /28 = 14 hosts
/29 = 6 hosts /30 = 2 hosts (PtP) /31 = 2 hosts (PtP)

SUBNETTING TOOLS



ipcalc
Calculate network, broadcast, host range, and more.



Online Calculators
e.g., subnetcalc.com, ipcalc.app



Spreadsheet
Build custom subnet plans and documentation.

BEST PRACTICES

- ✓ Plan your IP scheme before deployment.
- ✓ Use the smallest subnet that meets requirements.
- ✓ Document subnets, VLANs, and gateway addresses.
- ✓ Use VLSM to conserve address space.
- ✓ Summarize routes to reduce routing table size.

ROUTE SUMMARIZATION EXAMPLE

Instead of routing 4 subnets:

• 192.168.1.0/26
• 192.168.1.64/26
• 192.168.1.128/26
• 192.168.1.192/26

Use summary route:
192.168.1.0/24

Fewer routes = better performance!

CYBERSECURITY RELEVANCE



Smaller subnets limit broadcast traffic and attack surface.



Network segmentation improves security and access control.



Improper subnetting can cause IP conflicts and outages.



Understand subnetting to configure firewalls, ACLs, and routing securely.



KEY TAKEAWAY:

Subnetting is essential for building **scalable**, **efficient**, and **secure** networks.
Master subnetting to optimize **IP usage** and strengthen **network design**.

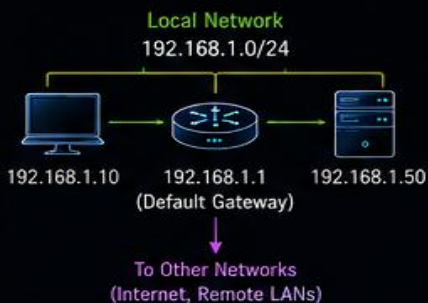
16. DEFAULT GATEWAY – THE EXIT DOOR

16

The default gateway is the next-hop router that forwards traffic from a local network to remote networks (outside the local subnet).

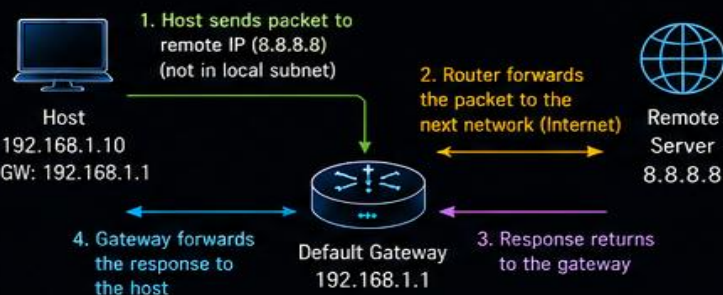
WHAT IS A DEFAULT GATEWAY?

- ✓ Host uses it to reach destinations outside its local subnet.
- ✓ Usually the IP address of the router interface on the local network.
- ✓ Only one default gateway per host, but can be changed.
- ✓ Without a default gateway, the host can only reach local subnet devices.



i If the destination IP is not in the local subnet, the packet is sent to the default gateway for routing to the next network.

HOW IT WORKS (EXAMPLE)



i The host does not know the full path. It only knows the default gateway. The router handles the rest.

CHECKING DEFAULT GATEWAY

Windows

```
C:\> ipconfig
```

```
Windows IP Configuration
...
Default Gateway . . . . : 192.168.1.1
```

Linux/macOS

```
$ ip route | grep default
```

```
default via 192.168.1.1 dev eth0
192.168.1.0/24 dev eth0 proto kernel ...
```

Example Output

Interface	IP Address	Subnet Mask	Default Gateway
Ethernet0	192.168.1.10	255.255.255.0	192.168.1.1

i The default gateway is the router interface IP in your subnet.

DEFAULT GATEWAY IN ACTION



Source	Destination	In Same Subnet?	Next Hop
192.168.1.10	192.168.1.50	✓ Yes	Directly (ARP)
192.168.1.10	10.0.0.20	✗ No	Default Gateway (192.168.1.1)
192.168.1.10	8.8.8.8	✗ No	Default Gateway (192.168.1.1)

i If the destination is outside the local subnet, traffic is sent to the default gateway.

STATIC vs DYNAMIC DEFAULT GATEWAY

Static (Manually Configured)

- ✓ Set manually on each host.
- ✓ Used in small, simple networks.
- ✓ Does not change unless reconfigured.
- ✓ Example: 192.168.1.1

Dynamic (Via DHCP)

- ✓ Automatically provided by DHCP server (Option 3 – Router).
- ✓ Ideal for large networks.
- ✓ Easier to manage and maintain.
- ✓ Example: 192.168.1.1

DHCP OFFER (Example)

Option	Length	Value
3 (Router)	4	192.168.1.1
Leased IP: 192.168.1.10		Subnet Mask: 255.255.255.0
Default Gateway: 192.168.1.1		DNS Server: 8.8.8.8

i Clients use the gateway IP received via DHCP for all remote traffic.

TROUBLESHOOTING DEFAULT GATEWAY

Common Issues

- ✗ Incorrect gateway IP.
- ✗ Router interface down.
- ✗ No route to the Internet.
- ✗ Gateway not reachable.
- ✗ Conflicting IP or subnet mismatch.

How to Troubleshoot

- ✓ Check IP configuration (ipconfig / ip addr).
- ✓ Ping the gateway IP.
- ✓ Check router status and cables.
- ✓ Review routing table (route print / ip route).
- ✓ Verify subnet mask and gateway are correct.

Useful Commands

	Windows	Linux/macOS
Test gateway	ping <gateway-ip>	ping <gateway-ip>
Show route table	route print	ip route netstat -rn
View config	ipconfig /all	ip addr
Trace route	tracert <destination>	traceroute <destination>



KEY TAKEAWAY:

The default gateway is your network's **exit door**. It allows your device to reach beyond the local network. Configure it correctly and ensure it's **reachable** for reliable **connectivity** to the world outside.

17. DNS – THE INTERNET'S PHONEBOOK

17

DNS (Domain Name System) translates human-friendly domain names into IP addresses so devices can find and communicate with each other on the Internet.

WHAT IS DNS?

- ✓ Distributed, hierarchical naming system for the Internet.
- ✓ Translates domain names (e.g., `www.example.com`) to IP addresses (e.g., `93.184.216.34`).
- ✓ Runs over UDP port 53 (TCP port 53 used for zone transfers and large responses).
- ✓ Essential for browsing websites, sending email, and many applications.

i Without DNS, we would have to remember IP addresses for every site!

DNS QUERY TYPES

Type	Description
Recursive Query	Resolver returns the best answer it can (find the final answer or an error).
Iterative Query	Server returns the best referral it has (next step), not the final answer.
Forward Lookup	Domain name → IP address (A/AAAA).
Reverse Lookup	IP address → Domain name (PTR).

i Most clients use recursive queries via a resolver.

PUBLIC DNS SERVERS (EXAMPLES)

Provider	IPv4	IPv6
Google	8.8.8.8	2001:4860:4860::8888
Cloudflare	1.1.1.1	2606:4700:4700::1111
Quad9	9.9.9.9	2620:fe::fe
OpenDNS (Cisco)	208.67.222.222	2620:119:35::35
CleanBrowsing	185.228.168.9	2a0d:2a00:1::2

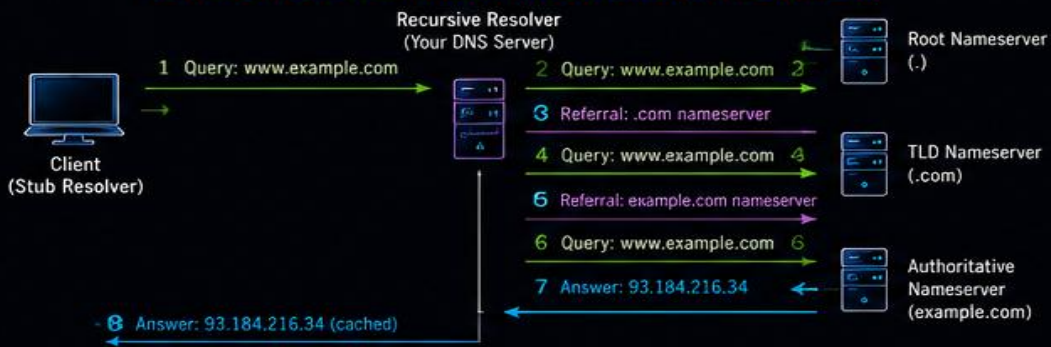
DNS SECURITY

- 🛡️ DNS Spoofing/Cache Poisoning: Attackers provide fake DNS responses.
- 🔍 Mitigation: Use DNSSEC, secure resolvers, and encrypted DNS.
- 🛡️ Encrypted DNS options:
 - DoT (DNS over TLS) = port 853
 - DoH (DNS over HTTPS) = port 443
- 🛡️ DNSSEC: Adds digital signatures to ensure authenticity and integrity.

USE CASES

- 🌐 Web Browsing: Resolves domain names to load websites.
- ✉️ Email Delivery: Uses MX records to route emails.
- ☁️ Application Connectivity: APIs, CDNs, and services rely on DNS.
- 🔗 Service Discovery (Internal): Internal apps, printers, and services.

HOW DNS RESOLUTION WORKS (RECURSIVE QUERY EXAMPLE)



i The resolver caches (stores) the answer for a period of time (TTL) to speed up future queries.

DNS MESSAGE FORMAT (OVERVIEW)

Transaction ID (16 bits)		Flags (16 bits)	
Flags (16 bits)	Questions (Count)	Bit	Meaning
Answer RRs (Count)	Authority RRs (Count)	QR (1)	0 = Query, 1 = Response
Additional RRs (Count)		Opcode (4)	Type of query
Questions Section (Variable)		AA (1)	Authoritative Answer
Answers Section (Variable)		TC (1)	Truncated Message
Authority Section (Variable)		RD (1)	Recursion Desired
Additional Section (Variable)		RA (1)	Recursion Available
		RCODE (4)	Response Code
Common RCODEs: 0 = No Error, 3 = NXDOMAIN, 2 = ServFail, 5 = Refused			

DNS RECORD TYPES (COMMON)

Type	Purpose	Example Value
A	Maps domain to IPv4 address	93.184.216.34
AAAA	Maps domain to IPv6 address	2606:2800:220:1:248:1893:25c8:1946
CNAME	Alias of another domain	<code>www.example.com</code> → <code>example.com</code>
MX	Mail exchange (email routing)	<code>mail.example.com</code> (priority 10)
NS	Authoritative name server	<code>ns1.example.com</code>
TXT	Text information (SPF, verification)	<code>"v=spf1 include:example.com ~all"</code>
PTR	Reverse DNS (IP to domain)	<code>34.216.184.93.in-addr.arpa</code>

DNS HIERARCHY



DNS CACHING & TTL

- ✓ Responses are cached for a time defined by TTL (Time To Live).
- ✓ Reduces query load and speeds up resolution.
- ✓ TTL values vary by record and domain administrator settings.
- ✓ Check cache with: `ipconfig /displaydns` (Windows) or `dig /nslookup`.
- i** Lower TTL = more updates, higher load. Higher TTL = faster, more cache efficiency.

COMMON DNS PORTS

Port	Protocol	Use
53	UDP	Standard DNS queries
53	TCP	Zone transfers, large responses, DNSSEC
5353	UDP	mDNS (local network) Avahi/Bonjour
853	TCP	DoT (DNS over TLS)
443	TCP/UDP	DoH (DNS over HTTPS)

TROUBLESHOOTING DNS ISSUES

- ✗ Can't resolve domain: Check internet, DNS server, and spelling.
- ✗ Slow DNS: Try another DNS server or flush cache.
- ✗ Wrong IP address: Could be cache issue or DNS spoofing.
- 🔧 Use tools: `nslookup`, `dig`, `ipconfig /flushdns`, `host`.
- ⚠️ Check firewall: Ensure port 53 is allowed.

KEY TAKEAWAY:

DNS is the Internet's phonebook. It **translates** names to IP addresses using a distributed system of servers. Understanding how DNS works helps you **troubleshoot** connectivity, improve **performance**, and enhance **security**.

18. DNS – BEYOND RESOLUTION

18

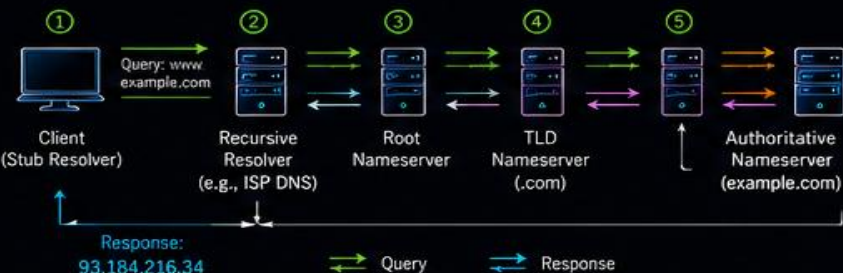
DNS is more than name resolution. It provides security, reliability, and efficiency for the modern Internet.

DNS ROLES

- ✓ Name Resolution – Maps names to IP addresses (A/AAAA).
- ✓ Service Discovery – Locates services (SRV, TXT, etc.).
- ✓ Load Distribution – Shares traffic across multiple servers.
- ✓ Security – DNSSEC, DNS over TLS/HTTPS to protect data.
- ✓ Policy & Control – Enforces access, filtering, and routing policies.

i DNS is a critical backbone service of the Internet.

DNS RESOLUTION FLOW (RECURSIVE)



i The resolver performs the work and returns the final answer to the client.

DNSSEC – ADDING TRUST

DNSSEC (DNS Security Extensions) ensures data authenticity and integrity.

1. Data is signed by the zone owner.
2. Signatures are stored in DNS records (RRSIG).
3. Resolver validates the signature using DNSKEY records.
4. Client receives verified, trusted data.

i DNSSEC prevents cache poisoning and spoofing attacks.

ENCRYPTING DNS TRAFFIC

Encryption protects DNS queries from eavesdropping and tampering.

Protocol	Port	Description	Use
Do53 (UDP)	53/UDP	Traditional DNS (not encrypted)	Legacy systems
DoT	853/TCP	DNS over TLS	Privacy, integrity
DoH	443/TCP	DNS over HTTPS	Bypass firewalls, browsers
DoQ	853/UDP	DNS over QUIC	Low latency, modern networks

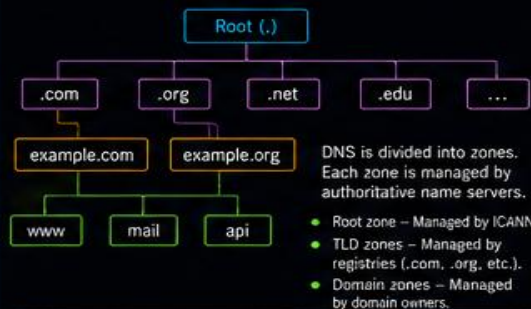
i DoH and DoT are widely used to secure DNS on public and enterprise networks.

DNS RECORDS FOR SERVICES

Record Type	Purpose	Example
A / AAAA	IPv4 / IPv6 address	www.example.com → 93.184.216.34
CNAME	Alias to another name	www → example.com
MX	Mail exchange (priority)	10 mail.example.com
NS	Name server delegation	ns1.example.com
SRV	Service location	_sip._tcp.example.com
TXT	Text information (SPF, DKIM)	"v=spf1 include:example.com ~all"
PTR	Reverse lookup (IP → name)	34.216.184.93.in-addr.arpa

i Different record types enable DNS to support many Internet services.

DNS ZONES



i Zones can be delegated to create a scalable DNS hierarchy.

DNS CACHING



Cache Table (Example)

Domain	Record	Value	TTL	Expires In
www.example.com	A	93.184.216.34	300	2:45
mail.example.com	A	93.184.216.35	600	8:12
example.com	NS	ns1.example.com	86400	22:15:30

i Caching reduces latency, server load, and improves performance.

COMMON DNS PROBLEMS & SOLUTIONS

Problem	Possible Cause / Solution
⚠ DNS name not resolving	Wrong DNS server / Check configuration, try another DNS.
🔄 Intermittent resolution	Low TTL / Network issues / Use reliable DNS.
⏱ Slow DNS responses	High latency / Use closer DNS servers, enable caching.
🛑 DNS poisoning / spoofing	No DNSSEC / Enable DNSSEC, use DoT/DoH.
🚫 Blocked DNS queries	Firewall / Try DoH (443) or different DNS server.

i Monitor DNS to ensure availability and performance.

DNS TOOLBOX

- > dig – Powerful DNS query tool.
- 🔍 nslookup – Simple DNS lookups.
- 📋 host – DNS lookup utility.
- 🛡 drill – DNSSEC-aware tool.
- 📊 dnsp perf – DNS performance testing.

BEST PRACTICES

- ✓ Use reliable, anycast DNS resolvers.
- ✓ Enable DNSSEC for zones.
- ✓ Use encrypted DNS (DoT/DoH) for privacy.
- ✓ Set appropriate TTL values.
- ✓ Monitor DNS performance and errors.
- ✓ Keep DNS infrastructure redundant.

MONITORING METRICS

- 🕒 Query Rate (QPS)
- 🕒 Response Time
- 🔄 Error Rate (SERVFAIL, NXDOMAIN)
- 📊 Cache Hit Ratio
- 🏠 Top Queried Domains
- 🛡 DNSSEC Validation Status

QUICK COMMANDS (EXAMPLES)

```
# Query A record
dig A www.example.com

# Query MX record
dig MX example.com

# DNSSEC validation
dig +dnssec example.com

# Reverse lookup
dig -x 93.184.216.34

# Trace resolution path
dig +trace www.example.com

# Query via DoT
dig @dns.example.com +tls-ca
```

CYBERSECURITY RELEVANCE

- 🛡 DNS is a common attack target.
- 🔒 Secure DNS protects users and data.
- 📋 Monitor and log DNS activity.
- 🔍 Use DNS filtering to block malicious domains.
- 👁 DNSSEC, DoT/DoH reduce risk of spoofing and eavesdropping.



KEY TAKEAWAY:

DNS is the Internet's **directory** service. Understanding its features, **security**, and best practices ensures reliable, **fast**, and **secure** communication in today's connected world.

19. DHCP – AUTOMATIC IP ADDRESSING

19

DHCP (Dynamic Host Configuration Protocol) automatically assigns IP address and network configuration to devices, reducing manual effort and IP address conflicts.

WHAT IS DHCP?

- ✓ Automatically assigns IP addresses and other network settings.
- ✓ Ensures efficient use of IP address space.
- ✓ Reduces manual configuration and errors.
- ✓ Supports dynamic IP allocation (leases).

i DHCP works on a client/server model using UDP.

HOW DHCP WORKS – DORA PROCESS



i Once complete, the client can use the IP address and other settings.

DHCP LEASE PROCESS

The assigned IP address is valid for a "lease time".

- ✓ 50% of lease time: Client tries to renew the lease with the same server.
- ✓ 87.5% of lease time: If renewal fails, client tries another DHCP server.
- ✓ Lease expires: IP address is released and can be assigned to another client.

Lease States	Description
INIT	Client has no IP address.
BOUND	Client has a valid IP address.
RENEWING	Client is trying to renew the lease.
REBINDING	Client is trying to rebind with any server.
EXPIRED	Lease expired, client has no IP.

DHCP MESSAGE DETAILS

Message	Direction	Description	UDP Port
DISCOVER	Client → Server	Finds available DHCP servers.	67 → 67 (Broadcast)
OFFER	Server → Client	Offers IP address and settings.	67 → 68 (Unicast/Bcast)
REQUEST	Client → Server	Requests the offered IP.	68 → 67 (Broadcast)
ACK	Server → Client	Confirms the IP and settings.	67 → 68 (Unicast)
NAK	Server → Client	Declines the request.	67 → 68
RELEASE	Client → Server	Releases the IP address.	68 → 67
DECLINE	Client → Server	IP is in use (conflict).	68 → 67
INFORM	Client → Server	Requests configuration only (no IP needed).	68 → 67

DHCP IP ADDRESS ALLOCATION METHODS

Method	Description	Use Case
Dynamic Allocation	IP is leased from a pool for a limited time.	Most common for clients.
Automatic Allocation	IP is always assigned from the same range.	Devices that need consistent IPs.
Manual Allocation	Specific IP is assigned based on MAC address.	Servers, printers, devices.

DHCP OPTIONS (COMMON)

Option Code	Option Name	Description	Example Value
1	Subnet Mask	Defines the subnet mask.	255.255.255.0
3	Router (Default Gateway)	Default gateway IP.	192.168.1.1
6	DNS Server	DNS server IP address.	8.8.8.8
15	Domain Name	DNS domain name.	example.com
51	Lease Time	IP lease duration (seconds).	86400 (1 day)
66	TFTP Server Name	TFTP server for boot files.	192.168.1.10
67	Boot File Name	Boot file name.	pxelinux.0

DHCP SERVER COMPONENTS

- IP Address Pool**
A range of IP addresses available for assignment.
- Scope**
Defines the network and parameters (pool, options, lease time).
- Lease Time**
Duration for which an IP address is valid.
- Reservations**
Binds a specific IP to a device (MAC).

DHCP RELAY AGENT

Used when DHCP server is on a different subnet.



- ✓ Forwards DHCP messages between client and server.
- ✓ Uses IP helper-address command (Cisco).

```
interface g0/0
ip helper-address 192.168.2.10
```

DHCP DISCOVER PACKET (EXAMPLE)

Source IP	0.0.0.0
Destination IP	255.255.255.255
Source Port	68 (Client)
Destination Port	67 (Server)
Message Type	DHCPDISCOVER
Client MAC	00:1A:2B:3C:4D:5E
Transaction ID	0x39AF3C12

i Uses broadcast to find any available DHCP server.



COMMON DHCP PORTS

Port	Protocol	Use
67	UDP	DHCP Server
68	UDP	DHCP Client
547	UDP	DHCPv6 Server
546	UDP	DHCPv6 Client

i DHCP uses UDP because it's simple and connectionless.

TROUBLESHOOTING DHCP

Problem	Possible Cause / Solution
⚠ No IP address	Check cable, DHCP server, scope, relay.
⚠ IP address conflict	Duplicate IPs, static IP clash, use DECLINE.
⚠ Can't reach network	Check default gateway, mask, routing.
⚠ Slow IP assignment	Check lease time, server load, scope size.

Useful Commands

```
Windows: ipconfig /all
Linux/macOS: ip addr, dhclient -v
Cisco: show ip dhcp binding, show ip dhcp pool
```

DHCPv6 (BASIC)

- ✓ Uses UDP ports 547 (server) and 546 (client).
- ✓ Messages: SOLICIT, ADVERTISE, REQUEST, REPLY.
- ✓ Supports Stateless (SLAAC) and Stateful DHCPv6.
- ✓ More efficient for IPv6 networks.



KEY TAKEAWAY:

DHCP simplifies IP management by **automatically** assigning and managing IP addresses and network settings. Understanding its process, **options**, and **troubleshooting** helps ensure a **reliable** and **efficient** network.

20. VLANS & NETWORK SEGMENTATION

VLANs (Virtual Local Area Networks) logically segment a physical network into multiple isolated broadcast domains to improve security, performance, and manageability.

WHAT IS A VLAN?

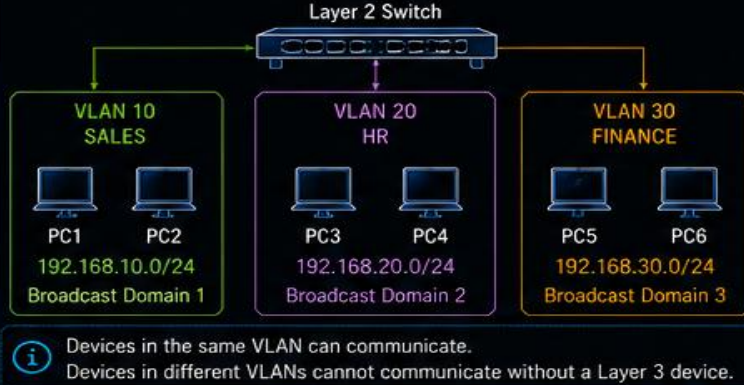
- ✓ A VLAN is a logical network that groups devices together regardless of their physical location.
- ✓ Devices in the same VLAN can communicate as if they are on the same LAN.
- ✓ VLANs create separate broadcast domains.

i VLAN ≠ Physical network
VLAN = Logical segmentation

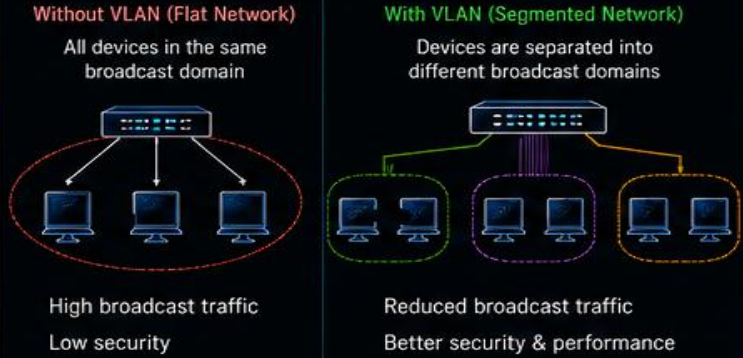
WHY USE VLANS?

- Improved Security**
Isolates sensitive departments (e.g., HR, Finance).
- Better Performance**
Reduces broadcast traffic.
- Simplified Management**
Easier to move users without changing configuration.
- Policy Control**
Apply different policies for different VLANs.

VLAN CONCEPT – SINGLE SWITCH EXAMPLE



VLAN vs PHYSICAL LAN



VLAN TAGGING – 802.1Q

When traffic goes over a trunk link, VLAN ID is added as a tag.



VLAN PORT TYPES

Port Type	Purpose	Carries	PVID	Example Use
Access Port	Connects end devices to a single VLAN	Untagged traffic	Yes	PC, Printer, IP Phone
Trunk Port	Connects network devices (switch to switch, switch to router)	Tagged traffic (multiple VLANs)	Yes	Switch-Switch, Switch-Router
Hybrid Port	Carries both tagged and untagged VLANs	Tagged + Untagged	Yes	Special use cases

i PVID (Port VLAN ID): The default VLAN for untagged frames on that port.

INTER-VLAN COMMUNICATION

Devices in different VLANs need a Layer 3 device (Router or Layer 3 Switch) to communicate.



SAMPLE VLAN CONFIGURATION (CISCO IOS)

1. Create VLANs

```
Switch(config)# vlan 10
Switch(config-vlan)# name SALES
Switch(config)# vlan 20
Switch(config-vlan)# name HR
Switch(config)# vlan 30
Switch(config-vlan)# name FINANCE
```

2. Assign Access Ports

```
Switch(config)# interface range fa0/1-2
Switch(config-if-range)# switchport mode access
Switch(config-if-range)# switchport access vlan 10
Switch(config)# interface range fa0/3-4
Switch(config-if-range)# switchport mode access
Switch(config-if-range)# switchport access vlan 20
```

3. Configure Trunk Port

```
Switch(config)# interface gi0/1
Switch(config-if)# switchport mode trunk
Switch(config-if)# switchport trunk allowed vlan 10,20,30
Switch(config-if)# no shutdown
```

BEST PRACTICES FOR VLANS

- ✓ Use VLANs to separate departments and sensitive data.
- ✓ Keep VLANs small to reduce broadcast traffic.
- ✓ Use meaningful VLAN IDs and names.
- ✓ Use ACLs between VLANs for fine-grained security.
- ✓ Document VLAN design and IP scheme properly.

COMMON VLAN IDS

VLAN ID	Name	Purpose (Example)
10	SALES	Sales Department
20	HR	Human Resources
30	FINANCE	Finance Department
40	GUEST	Guest Wi-Fi
99	MANAGEMENT	Network Management



KEY TAKEAWAY:

VLANs allow you to logically **segment** your network for **security**, **performance**, and **manageability**. Understanding VLANs is essential for building **scalable** and **enterprise-ready** networks.

21. FIREWALLS & ACLs – CONTROLLING TRAFFIC

21

Firewalls and Access Control Lists (ACLs) are the first line of defense in a network. They filter traffic based on rules to allow or deny access for security and performance.

WHAT IS A FIREWALL?

- ✓ A firewall is a security system that monitors and controls incoming and outgoing network traffic.
 - ✓ It creates a barrier between trusted (inside) and untrusted (outside) networks.
 - ✓ It allows legitimate traffic and blocks malicious or unauthorized traffic.
- i** Think of a firewall as a security guard who checks every packet against rules before allowing it in or out.

FIREWALL POSITION IN A NETWORK



HOW FIREWALLS WORK

- 1 Packet arrives at the firewall.
 - 2 Firewall checks the packet against its rule set.
 - 3 If a rule matches, the action (Allow/Deny) is applied.
 - 4 If no rule matches, default policy is applied (usually Deny).
 - 5 Allowed traffic is forwarded, denied traffic is dropped and logged.
- i** Rule order matters — the first match is applied.

TYPES OF FIREWALLS



Packet Filtering Firewall
Filters packets based on IP, port, protocol, and interface.



Stateful Inspection Firewall
Tracks the state of connections and allows only legitimate packets.



Next-Generation Firewall (NGFW)
Includes advanced features like IDS/IPS, app control, URL filtering, and more.

ACCESS CONTROL LIST (ACL) – OVERVIEW

- ✓ An ACL is a set of rules used by routers, switches, or firewalls to filter network traffic.
 - ✓ Rules can allow or deny traffic based on match criteria.
 - ✓ ACLs are processed top-down. The first matching rule is applied.
- i** If no rule matches, the implicit deny all rule is applied at the end.

ACL RULE STRUCTURE

Source IP Address	Source Wildcard	Destination IP Address	Destination Wildcard	Protocol	Port (Optional)	Action
192.168.1.0	0.0.0.255	10.0.0.0	0.255.255.255	TCP	80	Permit

i Wildcard mask is the inverse of subnet mask (e.g., 0.0.0.255 = /24).

SAMPLE ACL CONFIGURATION (CISCO IOS)

```
ip access-list extended BLOCK_WEBSITES
deny tcp any any eq 80
deny tcp any any eq 443
permit ip any any
```

Deny HTTP (port 80)
Deny HTTPS (port 443)
Allow all other traffic

i Apply the ACL to an interface using:

```
interface g0/0
ip access-group BLOCK_WEBSITES in
```

ACL PLACEMENT: INBOUND vs OUTBOUND

INBOUND

- ✓ Applied on an interface where traffic enters.
- ✓ Filters unwanted traffic as early as possible.



OUTBOUND

- ✓ Applied on an interface where traffic exits.
- ✓ Filters traffic leaving the network.



COMMON ACL MATCH CRITERIA

Criterion	Description
Source IP	IP address or network of the sender
Destination IP	IP address or network of the receiver
Protocol	TCP, UDP, ICMP, IP, etc.
Port Number	Specific port or range (e.g., eq 80, range 1024 65535)
Interface	Inbound or outbound interface
Time (Extended ACL)	Allow/Deny based on time range

BEST PRACTICES

- ✓ Deny what you don't need, allow what you do.
- ✓ Place ACLs close to the source of unwanted traffic.
- ✓ Keep ACLs simple and specific.
- ✓ Log denied traffic for monitoring and analysis.
- ✓ Review and update ACLs regularly.

COMMON USE CASES

- ✗ Block unnecessary ports (e.g., Telnet, FTP)
- ✗ Restrict access to internal servers
- ✓ Allow only required services (e.g., DNS, HTTP)
- ✗ Prevent unauthorized network access
- ✗ Segment and secure network zones



KEY TAKEAWAY:

Firewalls and ACLs help you **control**, **filter**, and **secure** network traffic. By applying the right rules at the right place, you **reduce risks**, prevent attacks, and ensure only **trusted traffic** flows through your network.

THANK YOU!

FOR YOUR TIME AND ATTENTION



SECURE



CONNECT



CONFIGURE



DEFEND



IMPROVE

Keep learning. Keep exploring. Keep building a better, safer digital world.
THE NETWORK IS YOURS TO MASTER.

STAY CURIOUS. STAY SECURE.